Jueun **Kim**

Ph.D. Student in CS, POSTECH

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EDUCATION

Ph.D., Computer Science and Engineering *Sep 2025 – Present*
Pohang University of Science and Technology (POSTECH), Pohang, South Korea
- Advisor: Prof. Wook-Shin Han

B.S., Computer Science and Engineering *Feb 2021 – Aug 2025*
Pohang University of Science and Technology (POSTECH), Pohang, South Korea
- Ranked 1st in the department
- Graduated Summa Cum Laude
- GPA: 4.04/4.3

Exchange Student, Computer Science *Sep 2024 – Feb 2025*
Technical University of Munich (TUM), Munich, Germany
- Achieved top grades in Deep Learning and NLP coursework

RESEARCH INTERESTS

Multi-hop and multi-modal information retrieval; retrieval for task-performing agents; long-context management; agent memory

EXPERIENCE

Intern *Jul 2024 – Aug 2024*
SK hynix
- Mobile FTL Team — developed an AI model predicting user commands to improve flash translation layer performance

Student Intern *Jun 2023 – Aug 2025*
Data Systems Lab, POSTECH
- Research Project I: proposed the Text-DB QA task and built its benchmark (ICLR 2026)
- Research Project II: developed an open-domain text-table QA model

PUBLICATIONS

CANOPY: Adaptive-Granularity Evidence Compression for Multimodal RAG *arXiv preprint arXiv:2610.00923*
Hyojeong Yun*, **Jueun Kim***, Wook-Shin Han (*equal contribution)

Hi-Q: Hierarchical Evidence-guided Query Refinement for Multi-Hop Question Answering *NeurIPS 2026*
Jueun Kim, Sungho Park, Wook-Shin Han

SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables *ICLR 2026*
Sungho Park, **Jueun Kim**, Wook-Shin Han

PROJECTS

CANOPY: Adaptive-Granularity Evidence Compression for Multimodal RAG *2026*
arXiv preprint arXiv:2610.00923
- **Problem:** Choosing a retrieval granularity does not settle how much of each retrieved item to keep: coarse units carry irrelevant content, uniformly fine selection strips the context needed to interpret the evidence, and existing compressors handle the trade-off per modality.
- **Contribution:** Represented retrieved items as hierarchies and scored regions against the query with a node encoder trained on gold evidence, selecting several granularities at once by parent-relative refinement with no LLM calls for node-level pruning; across five QA benchmarks over a 33M-item heterogeneous corpus, cut reader-input evidence tokens by 14.2–27.7% at comparable accuracy.

Hi-Q: Hierarchical Evidence-guided Query Refinement for Multi-Hop Question Answering *2026*
NeurIPS 2026
- **Problem:** A question is phrased at one granularity while the corpus is retrievable at another. Prior work imposes a fixed graph over the corpus, reformulates the query, or runs a generated program, but none decides when a query unit is already supported by evidence and when it must be refined.
- **Contribution:** Posed this as retrievable granularity discovery: a resolution operator tests each query node against retrieved evidence, resolved nodes terminate, and unresolved ones expand through a dependency-preserving binary operator checked by a semantic coverage verifier — 52.3 EM / 64.0 F1 averaged over three multi-hop QA benchmarks under full-corpus retrieval, 15.1 EM over IRCoT.

SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables *2026*
ICLR 2026

- **Problem:** Table-text multi-hop benchmarks are small and hand-curated, rarely going past two hops or requiring aggregation and grouping; generating them automatically instead raises whether a query is actually answerable and whether its phrasing reads naturally.
- **Contribution:** Built a fully automated SQL-centric pipeline that extracts atomic facts from passages into grounding tables and generates nested queries at a target hop count, with provenance-based refinement for executability and realistic-structure enforcement for phrasing — thousands of QA pairs at a quarter of HybridQA's annotation time, on which models scoring above 70 F1 on HybridQA and 50 F1 on OTT-QA lose more than 30 F1.

Home AI Agent with Event Memory *2026*
Korean government-hosted AI competition — finalist (top 20 of 1,100+ teams)

- **Problem:** Home cameras emit a stream of near-identical observations, so recovering what happened earlier means replaying footage, and recency-ordered logs bury the state changes that matter.
- **Contribution:** Built event memory that merges continuing scenes into events while preserving observation counts and state transitions — 72 observations into 6 events (91.7% fewer list entries), and cited the designated event in 6 of 7 past-event questions against 3 for a recent-3-records baseline.

Life-Event-Based Public Service AI Agent *2026*
Korean government-hosted AI competition — finalist (top 20 of 1,100+ teams)

- **Problem:** Public services are indexed by procedure name, so residents who only know their own situation cannot tell what to file; and because procedures span multiple steps and deadlines, an agent that does not retain their conditions and completed steps forces them to re-explain each time.
- **Contribution:** Kept the user's conditions and completion state as persistent memory and used it to decide what to surface next; linked 806 procedure items across 50 life events by 639 precedence relations, cutting precedence violations from 242 under deadline-only ordering to 0 on a 500-case regression.

TALKS

SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables *Jul 2026*
2026 Global AI Frontier Lab Workshop, Seoul

HONORS & AWARDS

POSTECHIAN Fellowship (Creative) *Nov 2025*
POSTECH

Grand Prize, SK hynix Internship Program *Aug 2024*
SK hynix

Popular Award, Undergraduate Research Project II *Jun 2024*
POSTECH Computer Science and Engineering Department

National Science & Technology Scholarship *Aug 2023*
KOSAF (Korea Student Aid Foundation)

Global Leadership Program Scholarship *Sep 2022*
POSTECH Computer Science and Engineering Department

TEACHING

Teaching Assistant — AI Application Specialist Course *2026*
Samsung Electronics

Teaching Assistant — Introduction to Data Analysis *2026*
POSTECH

Lecturer — POSTECHx Lecture, Pintos *2024*
POSTECH

Teaching Assistant — Data Programming Course *2023*
Samsung Electronics